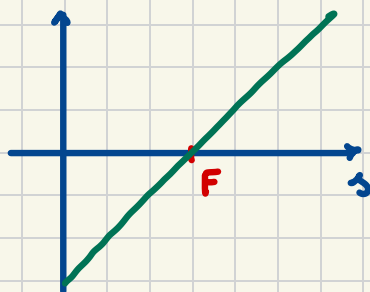


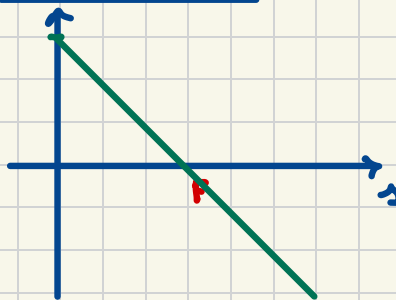
M3392: February 18th, 2025.

Review. An array of payoff curves.

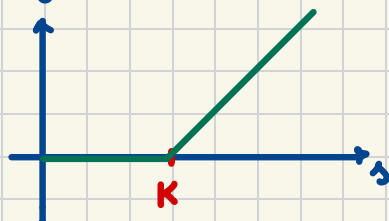
Long Forward. $S(T) - F$



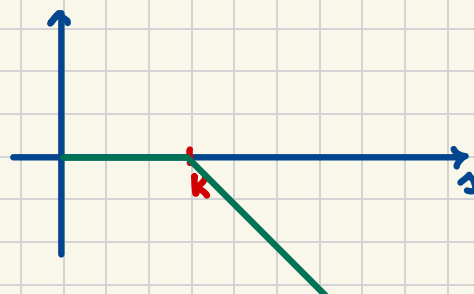
Short Forward. $F - S(T)$



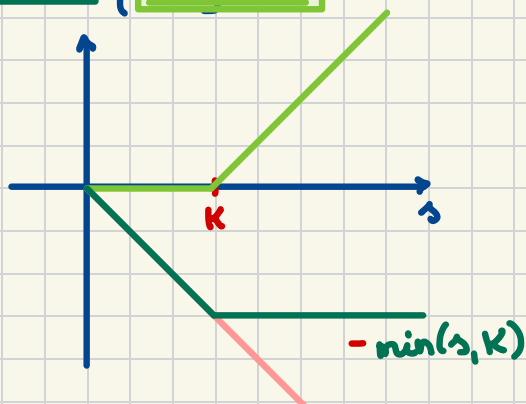
Long Call. $(S(T) - K)_+$



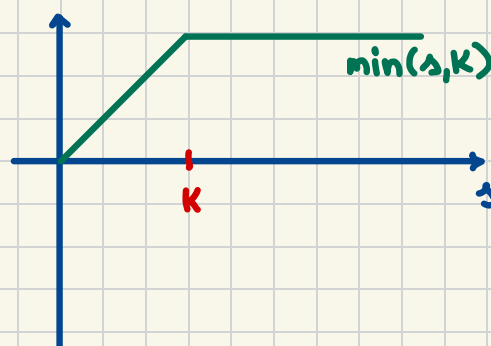
Short/Written Call. $-(S(T) - K)_+$



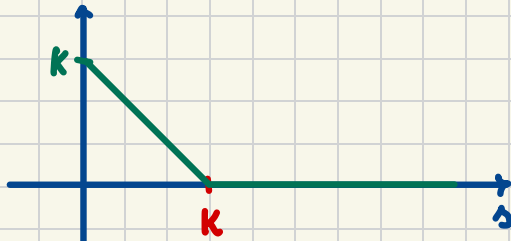
Cap. { Short Stock
Long Call }



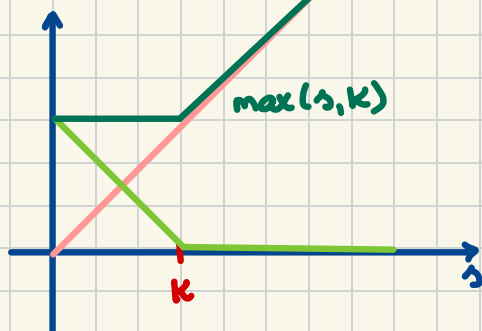
Covered Call. { Short Call
Long Stock. }



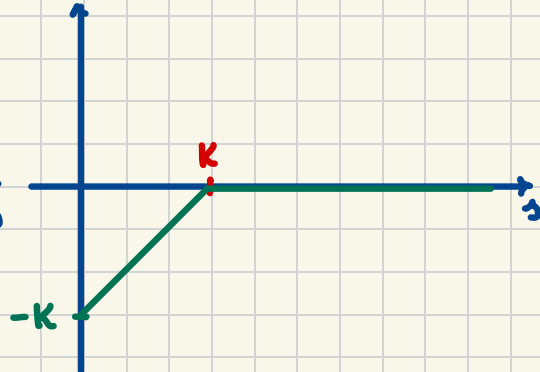
Long Put. $(K - S(T))_+$



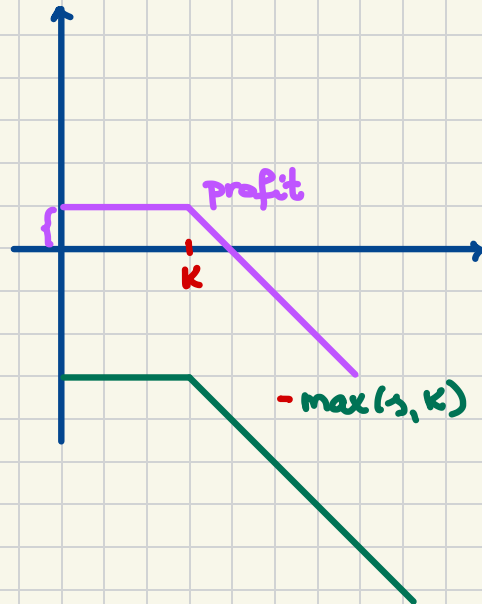
Floor. { Long Stock
Long Put



Short Put. $-(K - S(T))_+$



Covered Put { Short Put
Short Stock



Finite Probability Space.

... serve as environments for the possible paths that the asset price can take.

(e.g.)

$$S(T) \sim \begin{cases} 120 & \text{w/ prob. } \frac{1}{6} \\ 80 & \text{w/ prob. } \frac{1}{2} \\ 50 & \text{w/ prob. } \frac{1}{3} \end{cases}$$



Q: What is the expected payoff of a 105-strike put?

→:

$$V_p(T) = (K - S(T))_+$$

$$V_p(T) \sim \begin{cases} 0 & \text{w/ prob. } \frac{1}{6} \\ 25 & \text{w/ prob. } \frac{1}{2} \\ 55 & \text{w/ prob. } \frac{1}{3} \end{cases}$$

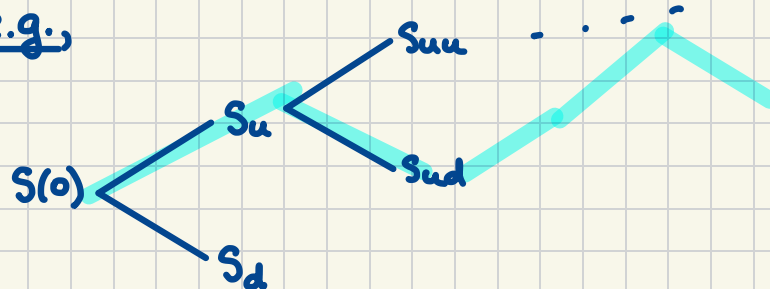
$$\mathbb{E}[V_p(T)] = 0 \cdot \frac{1}{6} + 25 \cdot \frac{1}{2} + 55 \cdot \frac{1}{3} = \dots$$



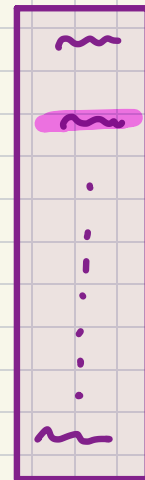
Caveat:

$$\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$$

e.g.)



All these finitely many scenarios are called **states of the world**.



We assume that

- each can happen, i.e., $\text{probab} > 0$
- and
- they exhaust all possibilities, i.e., $\sum \text{probab} = 1$

Arbitrage Portfolios.

Def'n. An **arbitrage portfolio** is a portfolio whose profit is

- nonnegative in ALL STATES OF THE WORLD, i.e.,
w/ PROBABILITY 1.
- and
- strictly positive in AT LEAST ONE state of the world,
i.e., w/ PROBABILITY > 0 .

Unless it's specified otherwise in a specific problem or example, we assume NO ARBITRAGE.